

Analyzing ILD BALF scRNA data

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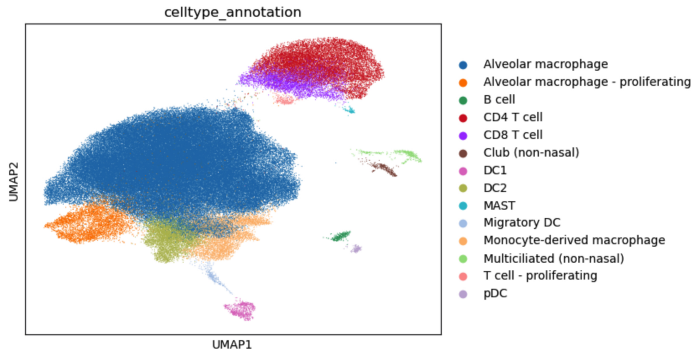
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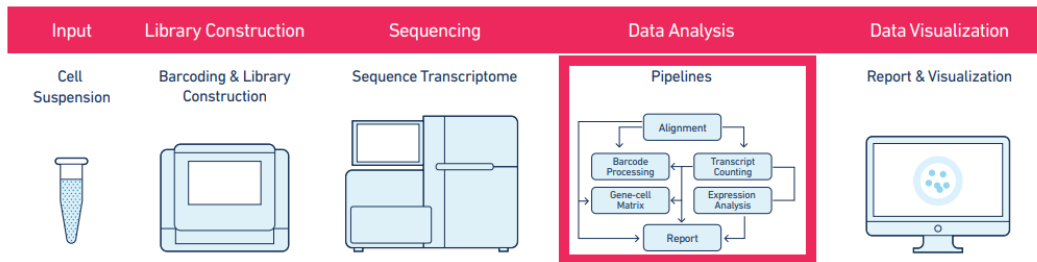


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scRNA - Ability to Contrast RNA profiles between cell-types

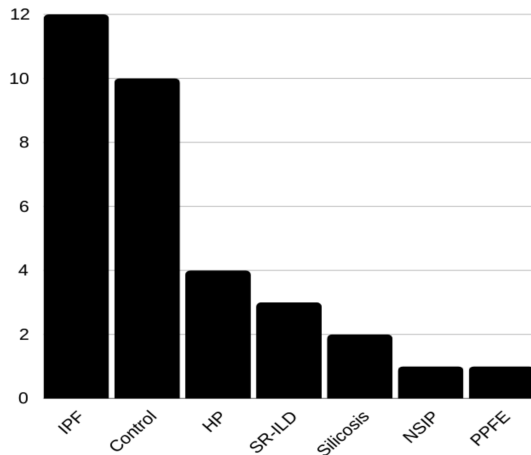
- Captures cellular heterogeneity often lost when performing bulk sequencing
- Additionally gives the power to detect sub-lineages within cell states not previously characterized





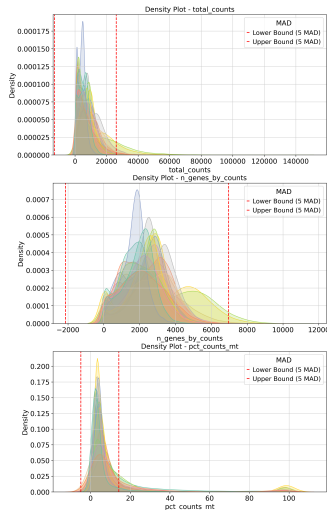
Study-specific Disease Demographics

- ILD scRNA data was generated for 33 BAL samples across 3 phases
- Includes 6 fibrosis-related ILDs



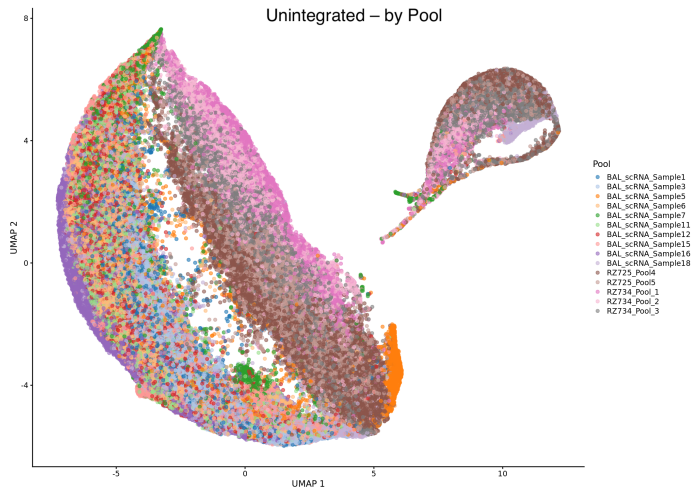
Cleaning and Pre-processing the data through Quality Control

- Assess data quality through metrics like total UMI counts per cell, how many genes within total counts and mitochondrial percentage
- Remove any ambient/background RNA
- Filter out low-quality cells and genes
- Check for batch effects



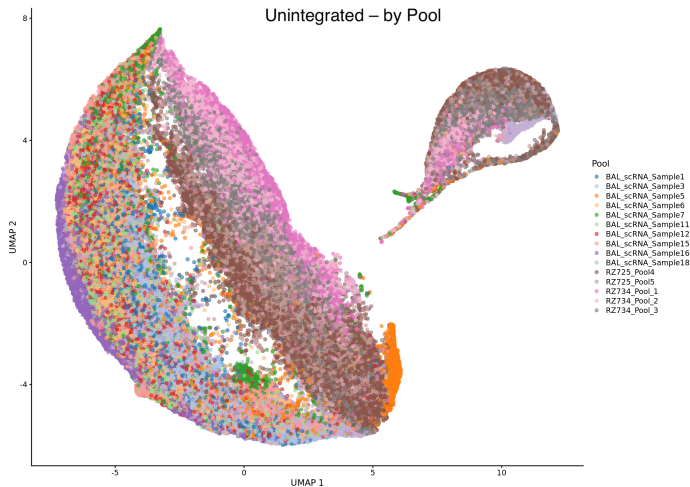
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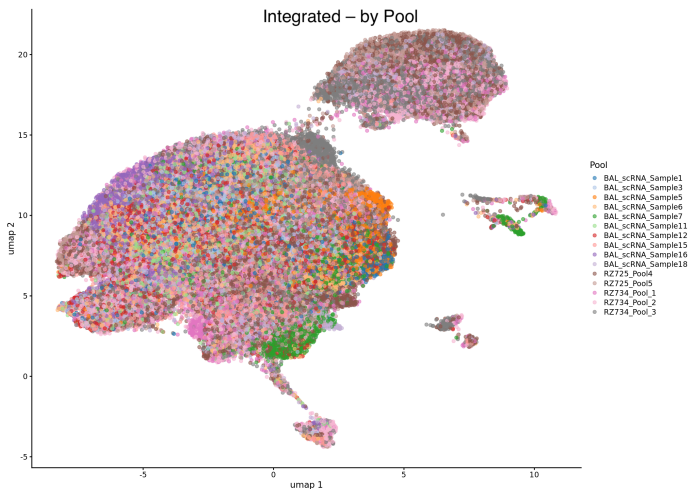
Integrating the data removes batch effects

- Integrating data from multiple samples allows for analyzing whether features are due to biology rather than an artefact of batch
- Ensures cells that are truly similar will be clustered and classified accurately
- Algorithms for integration:
 - scVI (🐍)
 - Harmony or Seurat (🔬)



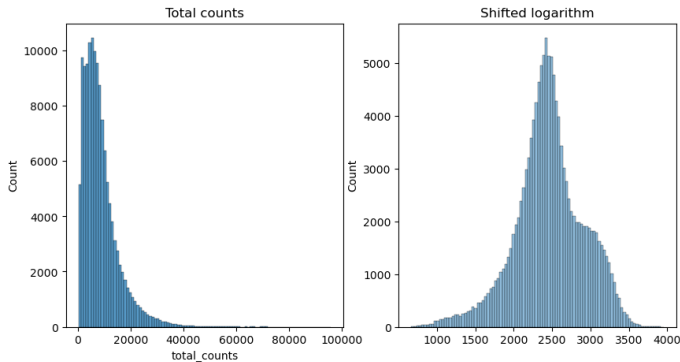
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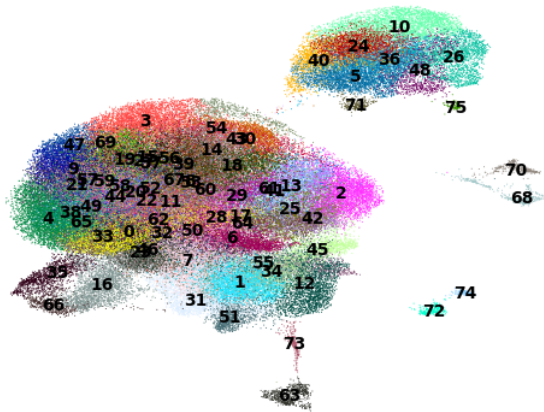
Normalizing Gene Expression

- Smooths differences due to technical variation
- Adjusts for cell-specific differences in RNA content for more robust gene expression comparison across cells



Clustering

- Group similar cells based on their gene expression profile
- Identify individual cell populations
- Clustering Methods:
 - Louvain
 - Leiden ★
 - K-means



- Annotation of clusters can be performed in two ways:
 - Unsupervised Machine Learning using reference data
 - Azimuth () / Celltypist ()
 - Scimilarity ()
 - Manual Annotation via known cell markers

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RESEARCH ARTICLE EISENBERG AND GIBBERDS

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Single-cell RNA sequencing reveals profibrotic roles of distinct epithelial and mesenchymal lineages in pulmonary fibrosis

ANDREW EISENBERG, AUSTIN GIBBERDS, LAUREN KAY, STEPHANIE VAND, MICHELLE WARDEN, CARAL ZALU, UNCHITR MEETHONG, DANIEL TAYLOR, J. J. WONG, JONATHAN A. KROEMER

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Annotating the clusters

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